Research Methodology

Week 1 Assignment: Questions, Answers and Explanations

Week 1: Group Discussion on Research

NPTEL course noc26_ge79 (SWAYAM), 10 questions, 1 point each

Correct answers are the NPTEL portal's accepted answers. For every option the table gives a verdict and the reason, and each question ends with a short explanation of the correct answer. Questions 7 to 10 are based on two reading passages, summarised in the shaded boxes.

Answer key at a glance

Question	Type	Correct option(s)	Your portal score
1	Select all that apply	**B, C**	0.5 (partial: only one of the two correct options chosen)
2	Single answer	**B**	1 (full marks)
3	Single answer	**D**	1 (full marks)
4	Single answer	**B**	1 (full marks)
5	Single answer	**A**	1 (full marks)
6	Single answer	**D**	1 (full marks)
7	Single answer	**B**	1 (full marks)
8	Single answer	**A**	1 (full marks)
9	Single answer	**B**	1 (full marks)
10	Single answer	**A**	1 (full marks)

Questions with explanations

Question 1 of 10

What sets a PhD apart from other degrees?

Select all that apply. *Your portal score: 0.5 (partial: only one of the two correct options chosen).*

Option	Statement	Verdict	Reason
A	Focus on practical skills and hands-on experience.	**Wrong**	Hands-on skill is built in many degrees (diplomas, engineering degrees), so it does not set a PhD apart.
B	**Development of critical thinking and analytical abilities.**	**Correct**	A PhD trains you to question, analyse and create new knowledge.
C	**Preparation to guide others in pursuing a PhD.**	**Correct**	Completing a PhD prepares you to supervise and guide future researchers.
D	Completion of coursework and examinations.	**Wrong**	Coursework and exams are the core of taught degrees; a PhD is judged on original research.

Correct answer: B and C.

Explanation: A PhD is about producing original knowledge. It builds critical thinking and analytical ability, and it prepares the holder to guide the next generation of researchers. Practical skills and coursework are common to other degrees as well, so they do not make a PhD different.

Question 2 of 10

What distinguishes a research-based degree from a typical undergraduate degree?

Single answer.

Option	Statement	Verdict	Reason
A	Clear and defined boundaries throughout the degree program.	**Wrong**	This describes the undergraduate degree, which has a fixed syllabus and end point. Research is the opposite.
B	**Opportunities for continuous discovery and learning.**	**Correct**	Research is open-ended: every result opens new questions to explore.
C	Graduation day marking the completion of the degree.	**Wrong**	Every degree ends with graduation, so this does not distinguish research.
D	Acquisition of expertise and recognition in a specific field.	**Wrong**	Other degrees also give expertise; it is a result, not what distinguishes research.

Correct answer: B.

Explanation: An undergraduate degree follows a fixed syllabus and ends at a known point. A research degree has no such boundary: you keep discovering and learning as new questions appear.

Question 3 of 10

What is the importance of attitude and communication in the student-advisor relationship?

Single answer.

Option	Statement	Verdict	Reason
A	Having prior knowledge is the most important factor for a student to work towards a PhD.	**Wrong**	Knowledge can be built during the PhD, so it is not the most important factor.
B	Advisors expect students to define their own research problems without any guidance.	**Wrong**	Advisors guide students; expecting no guidance misdescribes the relationship.
C	Students should choose advisors based on their reputation and availability.	**Wrong**	Reputation and availability ignore how well the two people can work together.
D	**Attitude, inclination to learn, and effective communication with the advisor are crucial.**	**Correct**	These let the advisor guide you and keep the work on track.

Correct answer: D.

Explanation: The relationship works when the student has a good attitude, wants to learn and communicates openly, so that the advisor can guide the work. The other options are either too extreme or ignore the working relationship.

Question 4 of 10

What makes research challenging for first-time researchers?

Single answer.

Option	Statement	Verdict	Reason
A	The need to handle routine work and drudgery.	**Wrong**	Routine work exists in research, but it is not what shocks a newcomer.
B	**The sudden shift in patterns from coursework to research.**	**Correct**	Coursework has structure and known answers; research has neither, and the abrupt change is hard.
C	The discovery process involved in understanding the research process.	**Wrong**	Discovery is the nature of research, not the specific challenge for first-timers.
D	The pressure to obtain a degree and complete the research process.	**Wrong**	Pressure applies to every researcher, not specifically to first-timers.

Correct answer: B.

Explanation: Students come from coursework, where tasks, deadlines and answers are defined. Research removes that structure all at once, and adjusting to this sudden change is the main difficulty for a first-time researcher.

Question 5 of 10

What is one advantage of research when dealing with failure?

Single answer.

Option	Statement	Verdict	Reason
A	**Research allows for changing initial assumptions and exploring different approaches.**	Correct	A failed attempt becomes information: you revise assumptions and try another route.
B	Research ensures that failures are minimized and success is guaranteed.	Wrong	No research can guarantee success.
C	Research provides a predefined path to follow, for sure eliminating the possibility of failure.	Wrong	Research has no predefined path, and failure cannot be eliminated.
D	Research discourages any changes in assumptions or approaches to maintain consistency.	Wrong	Research welcomes changes in assumptions; discouraging them would make failure fatal.

Correct answer: A.

Explanation: Research is flexible. When something fails you can change your assumptions or method and continue, so failure becomes a source of learning instead of an end point.

Question 6 of 10

When conducting experiments and analysing results in research:

Single answer.

Option	Statement	Verdict	Reason
A	Each individual step will always lead to something significant.	Wrong	"Always" is too strong; many single steps give nothing significant.
B	The familiarity with the area helps in building experiments, but the results are often insignificant.	Wrong	Calling the results "often insignificant" ignores that small results add up.
C	Day-to-day experiments and results are usually significant and groundbreaking.	Wrong	Most day-to-day results are modest, not groundbreaking.
D	**Small steps are taken, and over time they contribute to something significant.**	Correct	Progress comes from many small steps whose combined effect is significant.

Correct answer: D.

Explanation: Research advances through many small steps. Each step alone may look minor, but together they build something significant, so the other options overstate or understate how single steps matter.

Question 7 of 10

Passage A: Engineers and Scientists (summary)

The passage says gadgets now become outdated in two to three years even though they last physically much longer; storage media show this (gramophone records and cassettes lasted decades, while DVDs disappeared quickly). An industry engineer solves troubleshooting problems on tight timelines, with questions such as whether a lighter material can replace the current one or whether the product can work with fewer parts. A university scientist asks why a material property has its value and what its limits are.

Technologies in use today have life span of decades.

Single answer.

Option	Statement	Verdict	Reason
A	True	**Wrong**	The passage says phones, computers and gadgets become obsolete in two to three years.
B	**False**	**Correct**	Lifespans are short: even storage media such as DVDs disappeared quickly compared with older records and cassettes.

Correct answer: B.

Explanation: The passage stresses that the speed at which technologies appear and disappear is increasing, and that gadgets become outdated within two to three years. So their useful life is short, not decades.

Question 8 of 10

Based on Passage A (see Question 7).

Engineers aim to reduce the number of parts in a product.

Single answer.

Option	Statement	Verdict	Reason
A	**True**	**Correct**	One of the engineer's typical questions is whether the product can function with fewer parts than it has now.
B	False	**Wrong**	This contradicts the engineer's question listed in the passage.

Correct answer: A.

Explanation: Fewer parts lower the cost and raise the reliability of a product, which is why the passage lists "can it work with fewer parts?" among the engineer's questions.

Question 9 of 10

Passage B: Global warming (summary)

The passage says the planet's average temperature is rising. It gives very strong experimental evidence: 10 of the hottest recorded years fall in the last 15 or so, and temperature and atmospheric carbon dioxide are strongly correlated over the past 60 years and in 100,000 years of Antarctic ice-core data. Other gases also cause warming, but carbon dioxide is the largest contributor by quantity. The controversy is about the short-term economic cost of acting, not about whether warming is real.

Global warming is only a theoretical phenomenon.

Single answer.

Option	Statement	Verdict	Reason
A	True	**Wrong**	The passage gives measured, experimental evidence, so it is not merely theoretical.
B	**False**	**Correct**	Hottest-year records and temperature and CO2 data from direct measurements and ice cores show it is real.

Correct answer: B.

Explanation: A phenomenon backed by measured data (the hottest years on record and long temperature and carbon dioxide records) is observed and real, not only a theoretical idea.

Question 10 of 10

Based on Passage B (see Question 9).

Carbon dioxide in the atmosphere contributes to global warming.

Single answer.

Option	Statement	Verdict	Reason
A	**True**	**Correct**	The passage links atmospheric CO2 directly to the temperature rise and calls it the biggest contributor by quantity.
B	False	**Wrong**	This contradicts the strong correlation reported in the 60-year and 100,000-year data.

Correct answer: A.

Explanation: The passage says carbon dioxide directly affects the temperature rise, supported by the strong correlation between the two in recent and ice-core records, and it names carbon dioxide as the most significant contributor by quantity.